Retail Sales Data Analysis

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Introduction

This project focuses on performing Exploratory Data Analysis (EDA) on retail sales data to identify patterns, trends, and customer behavior. The goal is to extract meaningful insights that can help in better business decision-making.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import pandas as pd

df = pd.read_csv("retail_sales_dataset.csv")

df.head()
```

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
0	1	24-11-2023	CUST001	Male	34	Beauty	3	50	150
1	2	27-02-2023	CUST002	Female	26	Clothing	2	500	1000
2	3	13-01-2023	CUST003	Male	50	Electronics	1	30	30
3	4	21-05-2023	CUST004	Male	37	Clothing	1	500	500
4	5	06-05-2023	CUST005	Male	30	Beauty	2	50	100

Next steps: Generate code with df New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Transaction ID    1000 non-null   int64
 1   Date              1000 non-null   object
 2   Customer ID       1000 non-null   object
 3   Gender            1000 non-null   object
 4   Age               1000 non-null   int64
 5   Product Category  1000 non-null   object
 6   Quantity          1000 non-null   int64
 7   Price per Unit    1000 non-null   int64
 8   Total Amount      1000 non-null   int64
dtypes: int64(5), object(4)
memory usage: 70.4+ KB
```

```
df.describe()
```

	Transaction ID	Age	Quantity	Price per Unit	Total Amount
count	1000.000000	1000.00000	1000.000000	1000.000000	1000.000000
mean	500.500000	41.39200	2.514000	179.890000	456.000000
std	288.819436	13.68143	1.132734	189.681356	559.997632
min	1.000000	18.00000	1.000000	25.000000	25.000000
25%	250.750000	29.00000	1.000000	30.000000	60.000000
50%	500.500000	42.00000	3.000000	50.000000	135.000000
75%	750.250000	53.00000	4.000000	300.000000	900.000000
max	1000.000000	64.00000	4.000000	500.000000	2000.000000

```
df.isnull().sum()
```

	0
Transaction ID	0
Date	0
Customer ID	0
Gender	0
Age	0
Product Category	0
Quantity	0
Price per Unit	0
Total Amount	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.columns
```

```
Index(['Transaction ID', 'Date', 'Customer ID', 'Gender', 'Age',
       'Product Category', 'Quantity', 'Price per Unit', 'Total Amount'],
      dtype='object')
```

```
df['Total Amount'].mean()
```

```
np.float64(456.0)
```

```
df['Total Amount'].median()
```

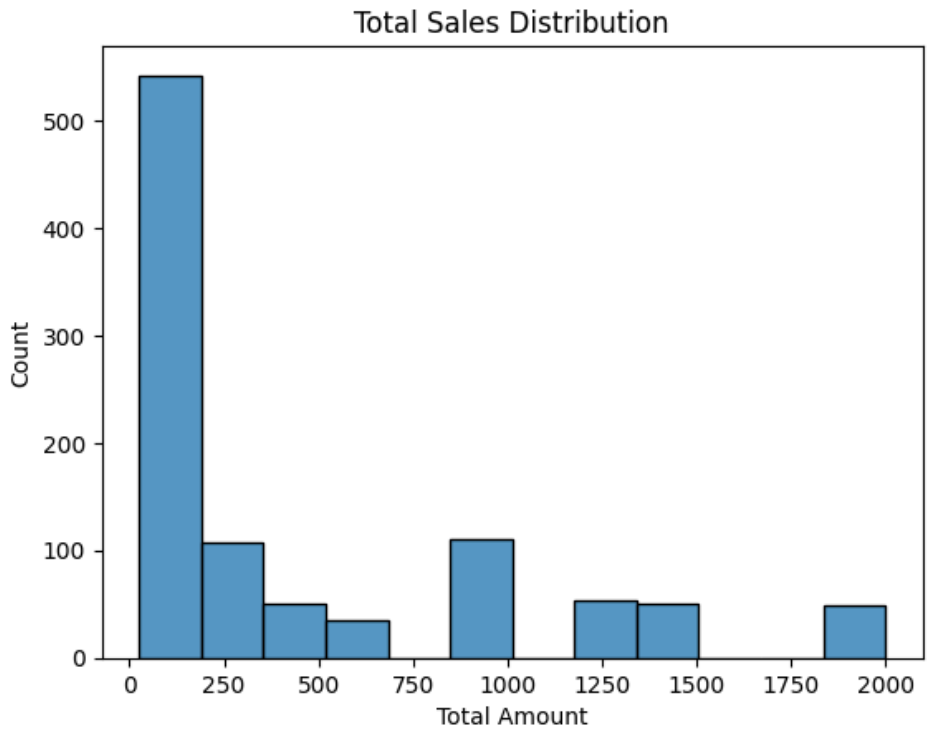
```
135.0
```

```
df['Total Amount'].std()
```

```
559.997631555123
```

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
import matplotlib.pyplot as plt

sns.histplot(df['Total Amount'])
plt.title("Total Sales Distribution")
plt.show()
```

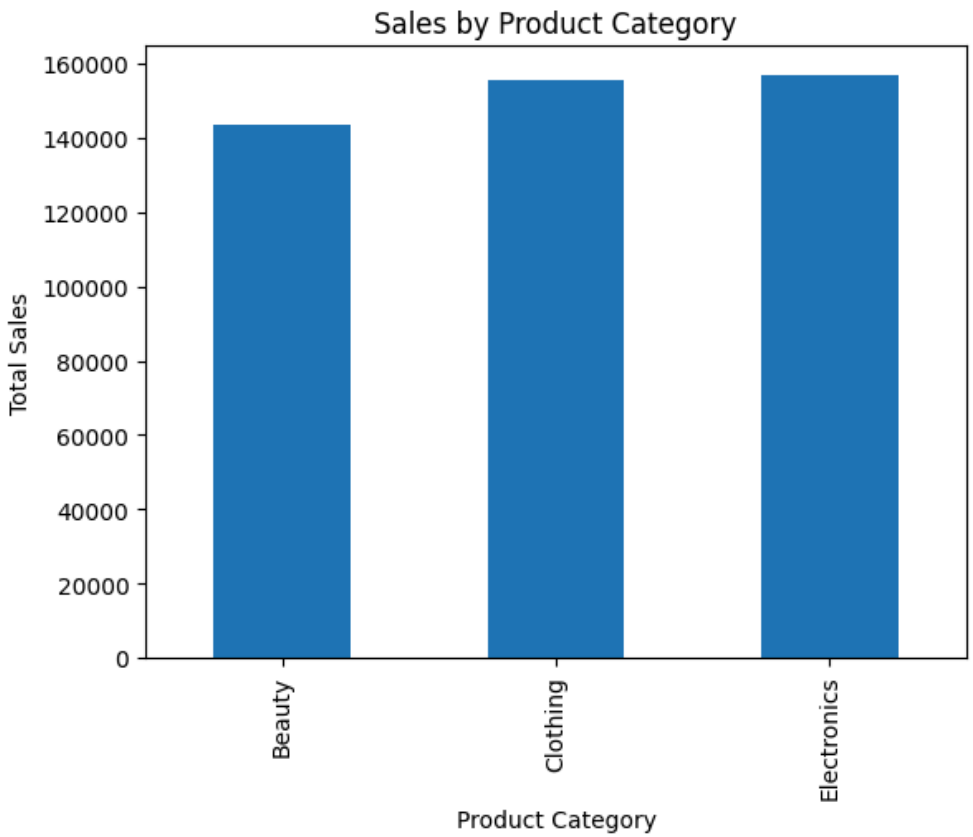


Total Sales Distribution

Insight:

The histogram of 'Total Sales Distribution' shows a right-skewed distribution, indicating that the majority of transactions have lower total amounts, while a smaller number of transactions account for very high total sales. The distribution is characterized by a long tail extending towards higher values. This suggests that while most customers make smaller purchases, there are occasional high-value transactions that significantly contribute to the overall sales. Companies might consider strategies to encourage more frequent high-value purchases or to increase the average transaction value for the majority of customers.

```
df.groupby('Product Category')['Total Amount'].sum().plot(kind='bar')

plt.title("Sales by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Total Sales")
plt.show()
```



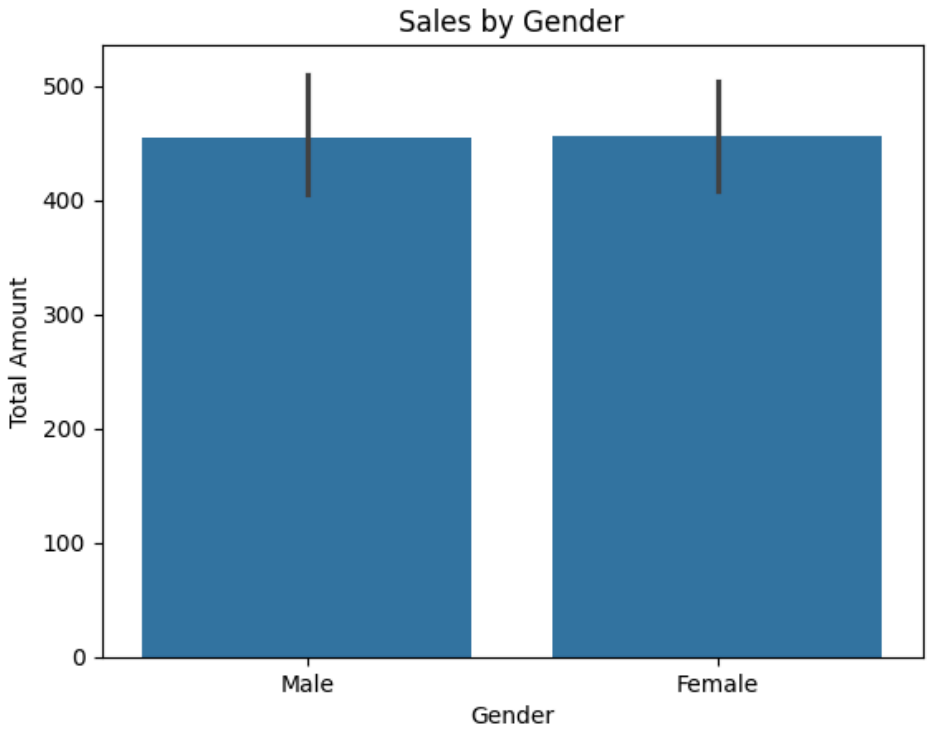
Sales By Product

Insight:

The chart shows that Electronics has the highest total sales, followed closely by Clothing, while Beauty has the lowest sales among the three categories. This suggests that customers purchase more electronics and clothing products compared to beauty items. The

company may focus on promoting beauty products to increase their sales.

```
sns.barplot(x='Gender', y='Total Amount', data=df)

plt.title("Sales by Gender")
plt.show()
```



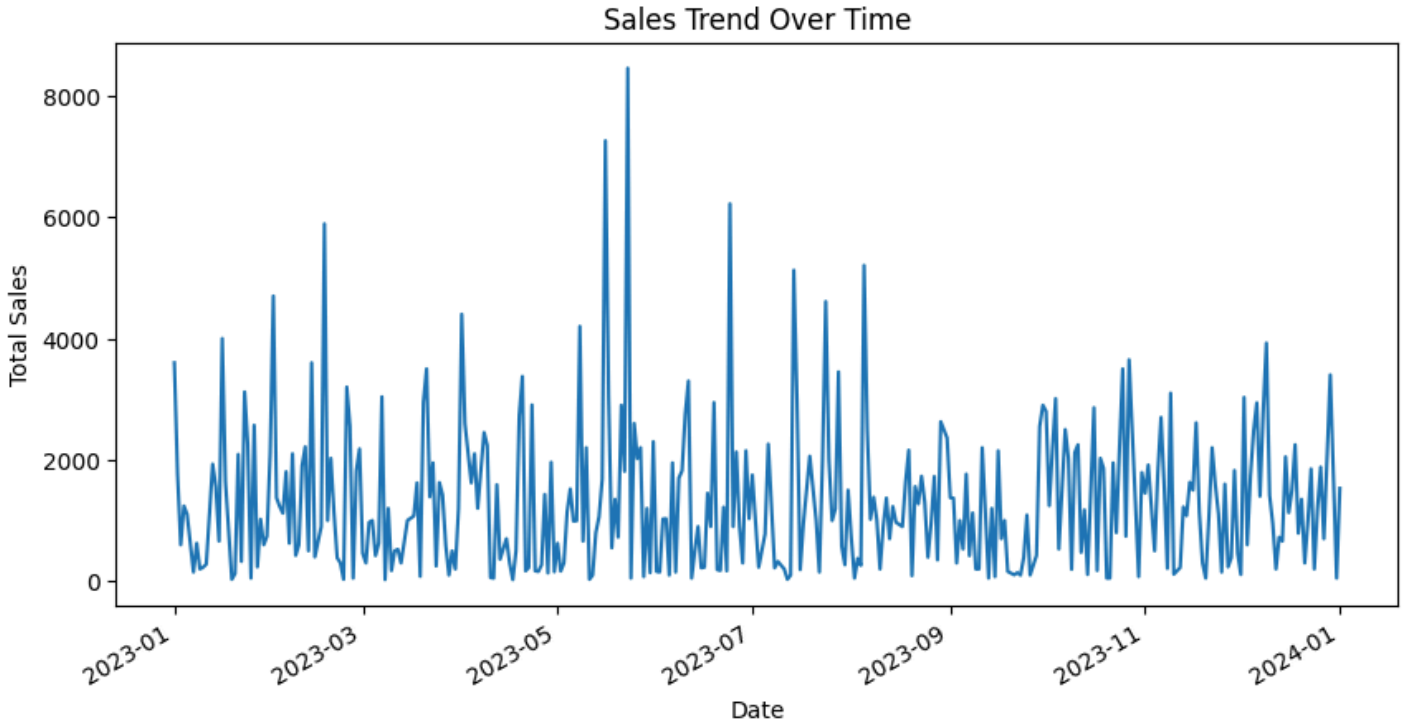
Sales By Gender

Insight:

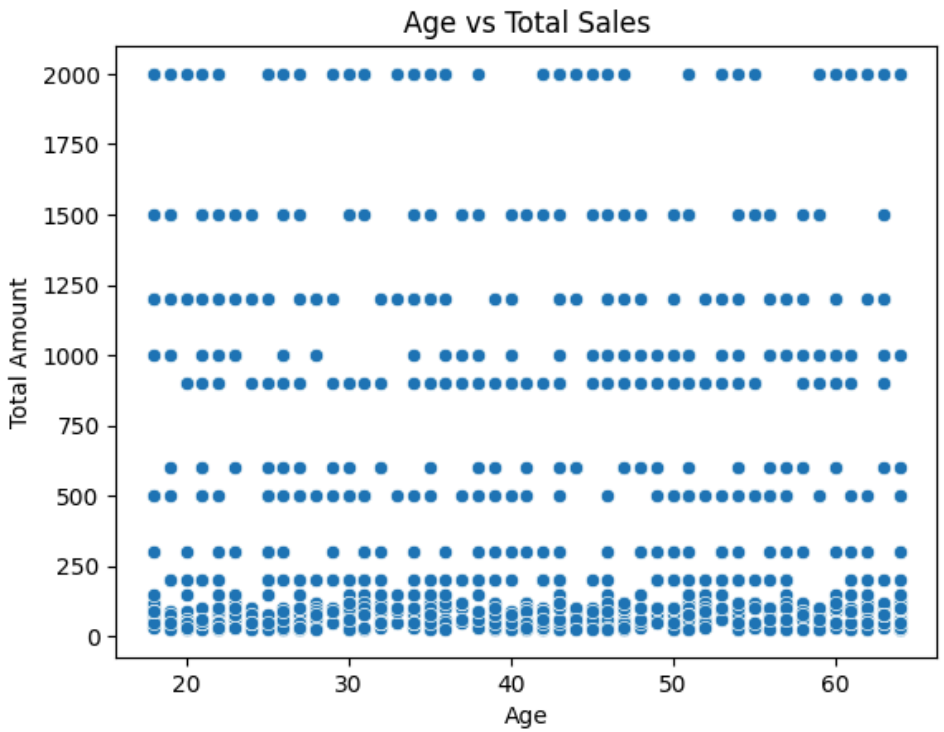
Sales from male and female customers are almost equal, with females contributing slightly more. This indicates no significant gender-based difference in purchasing behavior.

```
df['Date'] = pd.to_datetime(df['Date'])

sales_trend = df.groupby('Date')['Total Amount'].sum()

sales_trend.plot(figsize=(10,5))

plt.title("Sales Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Total Sales")
plt.show()
```

```
/tmp/ipykernel_148/3218615733.py:1: UserWarning: Parsing dates in %d-%m-%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=Tr
  df['Date'] = pd.to_datetime(df['Date'])
```




```
sns.scatterplot(x='Age', y='Total Amount', data=df)

plt.title("Age vs Total Sales")
plt.show()
```

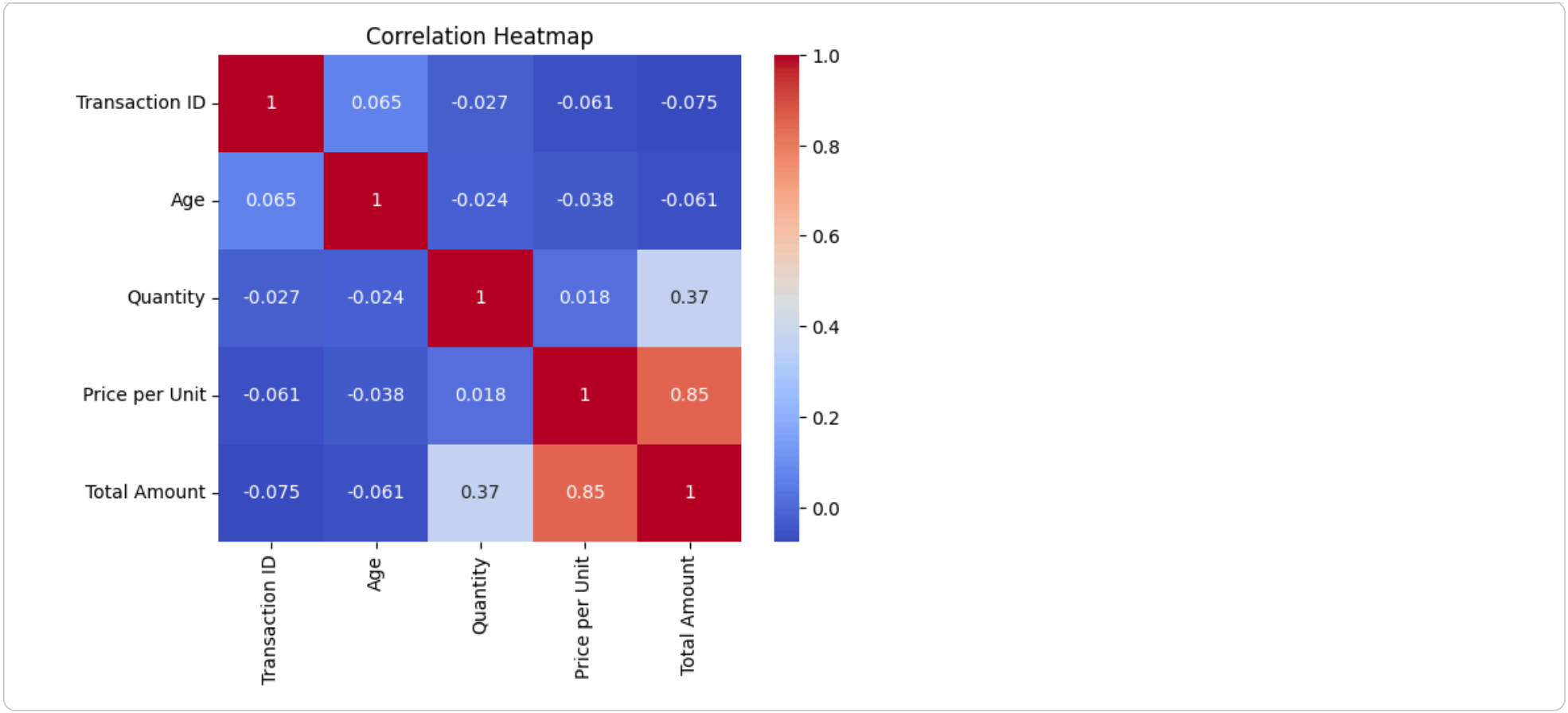


Age vs Total Sales Scatter Plot

Insight:

The scatter plot illustrates the relationship between customer age and total purchase amount. The points are widely distributed across different age groups, indicating no strong relationship between age and total sales. Customers from various age ranges appear to spend similar amounts, suggesting that purchasing behavior is relatively consistent across age groups.

```
corr = df.corr(numeric_only=True)

sns.heatmap(corr, annot=True, cmap="coolwarm")

plt.title("Correlation Heatmap")
plt.show()
```



Correlation Heatmap

Insight:

The correlation heatmap shows the relationships between numerical variables in the dataset. Price per Unit and Total Amount have a strong positive correlation (0.85), indicating that higher product prices significantly increase the total sales amount. Quantity and Total Amount show a moderate positive correlation (0.37), meaning purchasing more items also increases total sales. Other variables such as Age and Transaction ID show very weak correlations with sales, suggesting they have little direct impact on the total purchase value.

Conclusion

The exploratory data analysis of retail sales data reveals that most transactions involve lower purchase amounts, with a few high-value transactions contributing significantly to overall sales. Among product categories, electronics and clothing generate the highest revenue, while beauty products lag behind. Sales are almost equally distributed between male and female customers, showing no major gender-based difference. Additionally, factors like age do not have a strong influence on purchasing behavior. Price per unit plays a significant role in determining total sales, with quantity also contributing moderately.

Recommendations

- Focus on promoting beauty products to improve their relatively lower sales performance.
- Leverage high-performing categories like electronics and clothing through targeted marketing strategies.
- Introduce offers or discounts to encourage higher-value purchases and increase overall sales.
- Since gender and age do not significantly impact sales, marketing campaigns can target a broad audience.
- Optimize pricing strategies, as price per unit strongly influences total revenue.